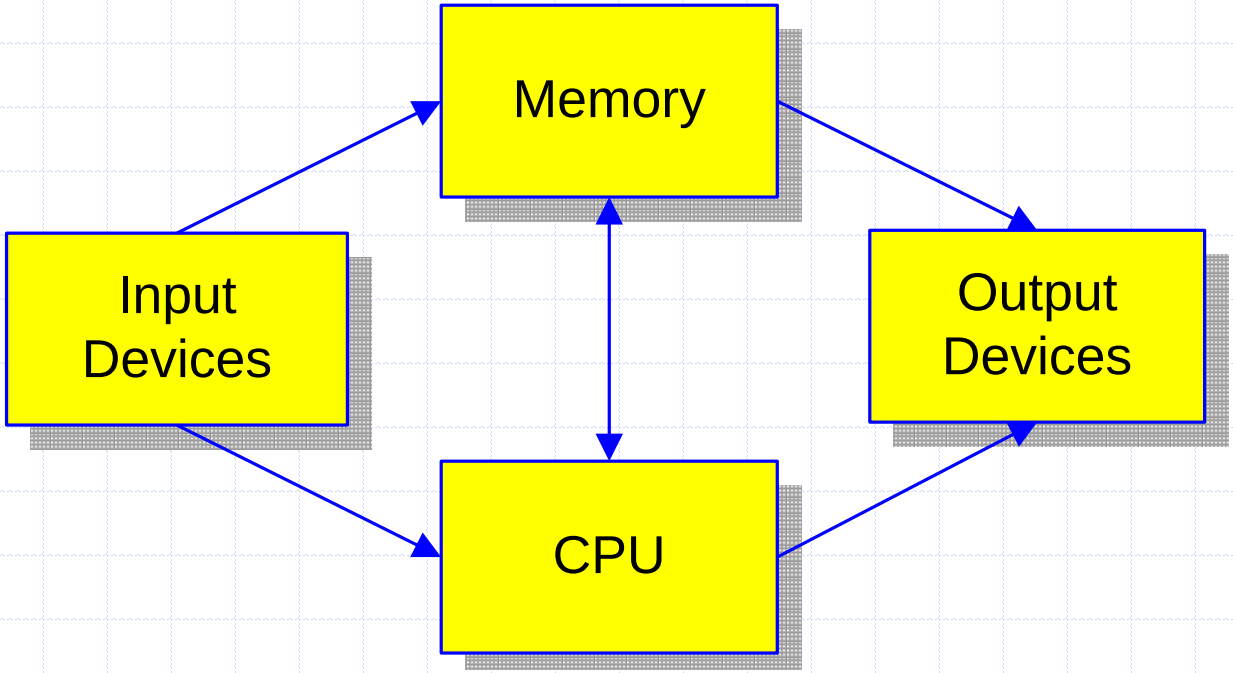
Basics

Machine, software, and program design

Computer Organization

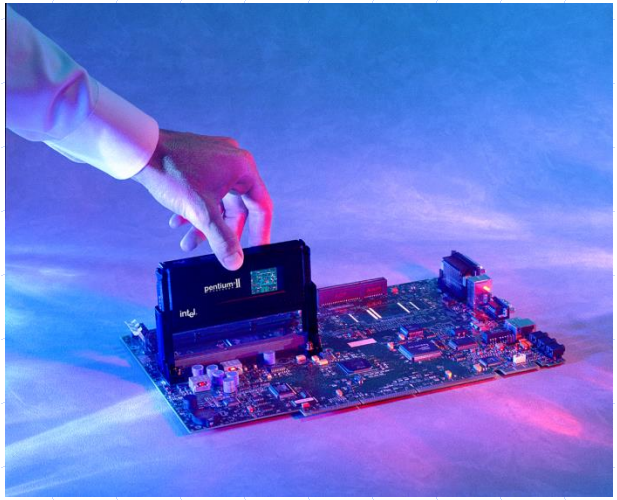
- CPU - central processing unit
  - Where decisions are made, computations are performed, and input/output requests are delegated
- Memory
  - Stores information to be processed by the CPU
- Input devices
  - Allows people to supply information to computers
- Output devices
  - Allows people to receive information from computers

Computer Organization



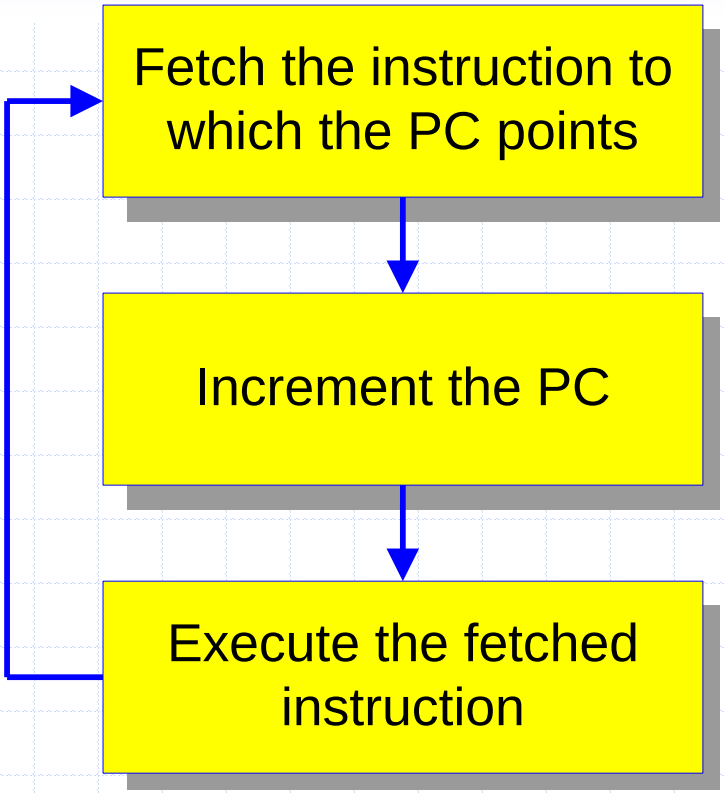
CPU

- *Brains* of the computer
  - Arithmetic calculations are performed using the Arithmetic/Logical Unit or ALU
  - Control unit decodes and executes instructions
- Arithmetic operations are performed using binary number system



Control Unit

- The fetch/execute cycle is the steps the CPU takes to execute an instruction
- Performing the action specified by an instruction is known as *executing the instruction*
- The program counter (PC) holds the memory address of the next instruction

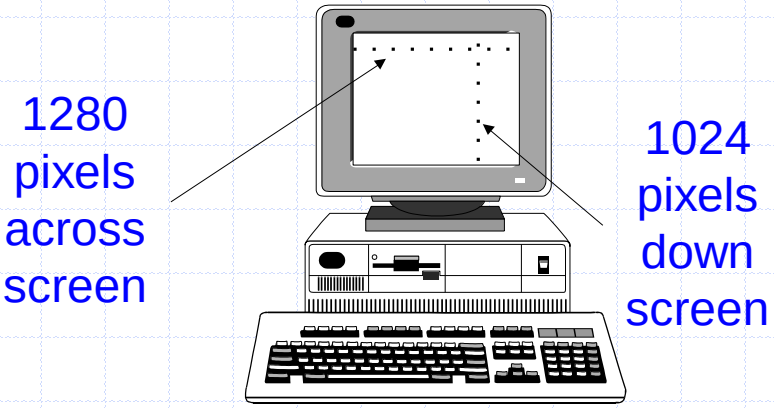


Input and Output Devices

- Accessories that allow computer to perform specific tasks
  - Receive information for processing
  - Return the results of processing
  - Store information
- Common input and output devices
  - Speakers
  - Printer
  - Keyboard
  - Mouse
  - Joystick
  - Microphone
  - Scanner
  - CD-ROM
  - DVD
- Some devices are capable of both input and output
  - Floppy drive
  - Hard drive
  - Magnetic tape units

Monitor

- Display device that operates like a television
  - Also known as CRT (cathode ray tube)
- Controlled by an output device called a *graphics card*
- Displayable area
  - Measured in dots per inch, dots are often referred to as pixels (short for picture element)
  - Standard resolution is 640 by 480
  - Many cards support resolution of 1280 by 1024 or better



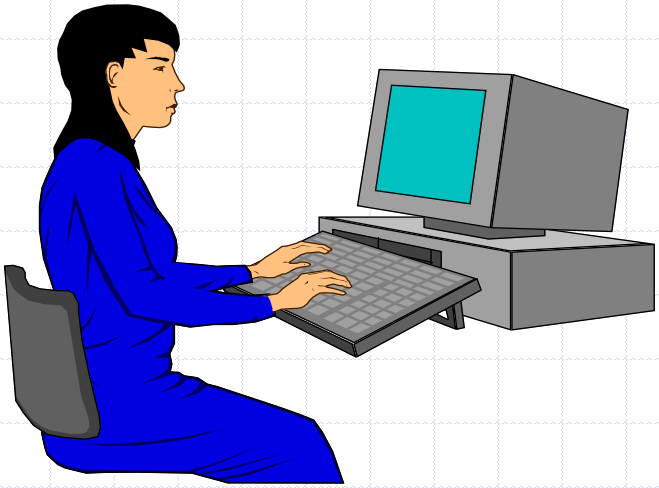
Software

- Application software
  - Programs designed to perform specific tasks that are transparent to the user
- System software
  - Programs that support the execution and development of other programs
  - Two major types
    - Operating systems
    - Translation systems

Application Software

- Application software is the software that has made using computers indispensable and popular

- Common application software
  - **Word processors**
  - Desktop publishing programs
  - Spreadsheets
  - Presentation managers
  - Drawing programs



- *Learning how to develop application software is our focus*

Operating System

- Examples
  - Windows®, UNIX®, Mac OS X®
- Controls and manages the computing resources
- Important services that an operating system provides
  - File system
    - Directories, folders, files
  - Commands that allow for manipulation of the file system
    - Sort, delete, copy
  - Ability to perform input and output on a variety of devices
  - Management of the running systems

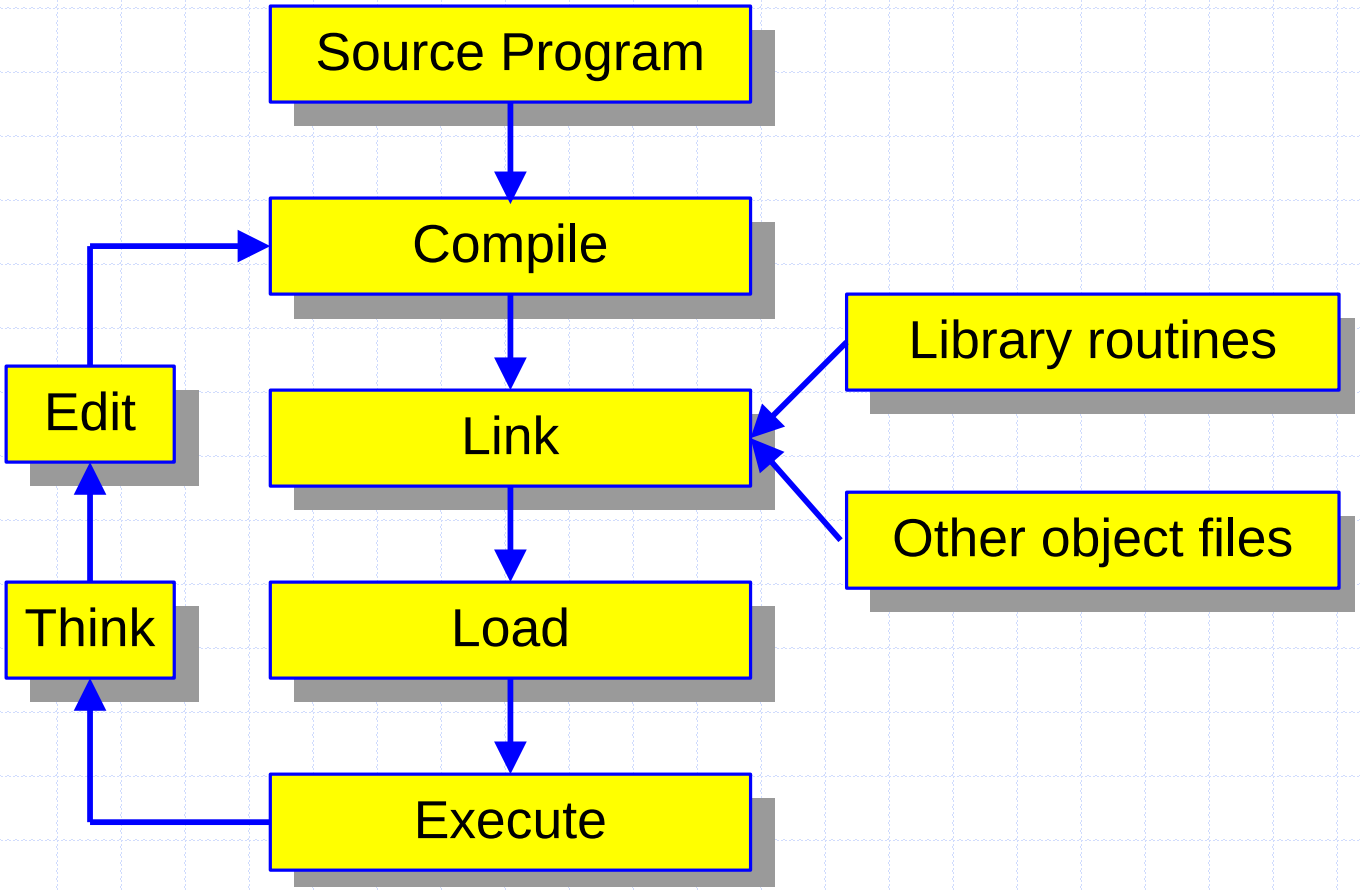
Translation System

- Set of programs used to develop software
- A key component of a translation system is a translator
- Some types of translators
  - Compiler
    - Converts from one language to another
  - Linker
    - Combines resources
- Examples
  - Microsoft Visual C++®, CBuilder®, g++, Code Warrior®
    - Performs compilation, linking, and other activities.

Software Development Activities

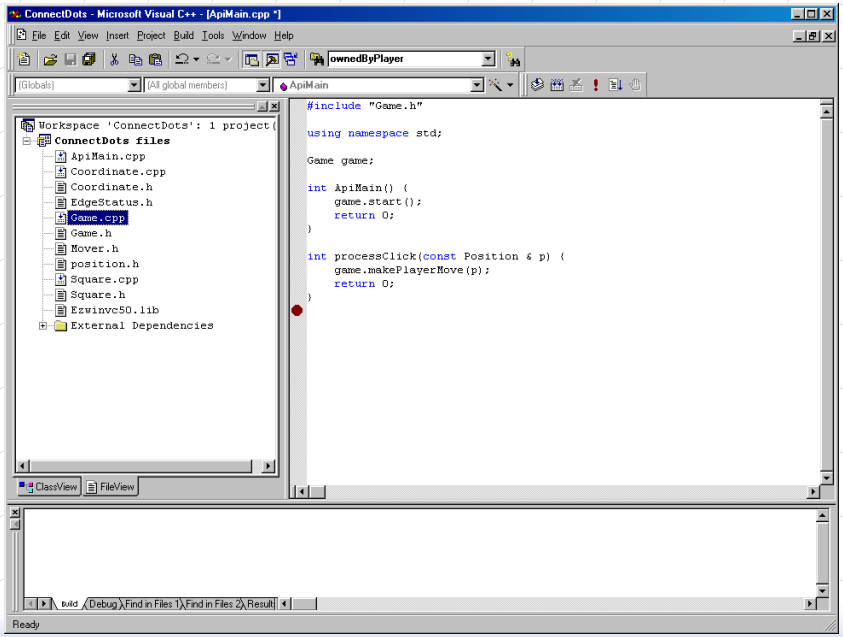
- Editing
- Compiling
- Linking with precompiled files
  - Object files
  - Library modules
- Loading and executing
- Viewing the behavior of the program

Software Development Cycle



IDEs

- Integrated Development Environments or IDEs
  - Supports the entire software development cycle
    - E.g., MS Visual C++, Borland, Code Warrior
- Provides all the capabilities for developing software
  - Editor
  - Compiler
  - Linker
  - Loader
  - Debugger
  - Viewer



What is Next ??

- Programming Fundamentals ..... !!!!